

## TOLBO, Mongolia

WMO: 442170

Lat: 48.417N

Lon: 90.300E

Elev: 6893

StdP: 11.39

Time Zone: 7.00 (E07)

Period: 02-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -30.5      | -26.3 | -35.5                         | 0.9 | -29.8 | -31.8 | 1.2 | -25.7 | 18.4                  | 11.4 | 15.4 | 8.4  | 0.4                   | 190  | 0.730 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 18.7                   | 72.8            | 53.2 | 70.2 | 51.7 | 67.5 | 50.5 | 56.0                | 68.0 | 54.2 | 65.7 | 52.5 | 63.9 | 6.5                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 51.4                            | 72.9 | 61.0 | 49.1 | 66.8 | 59.1 | 47.1 | 62.1 | 57.5 | 26.8          | 67.9 | 25.5 | 65.6 | 24.3 | 64.1 | 65.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |  |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|--|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |  |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |  |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |  |
| 18.2              | 15.5 | 13.4 | DB                         | -33.0 | 77.2               | 5.9 | 2.7                                                | -37.3 | 79.1       | -40.7 | 80.7       | -44.0 | 82.2       | -48.3 | 84.1 |  |
|                   |      |      | WB                         | -33.3 | 58.7               | 5.8 | 4.3                                                | -37.4 | 61.8       | -40.8 | 64.3       | -44.0 | 66.7       | -48.2 | 69.8 |  |

## Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 27.9   | -5.6  | 0.9   | 17.4  | 32.1  | 41.5  | 53.1  | 56.3  | 53.0  | 42.1  | 28.7  | 12.7  | 0.2   |
|                                                                   | DBStd       | 22.82  | 10.66 | 10.58 | 9.56  | 8.18  | 7.34  | 5.58  | 4.72  | 4.88  | 6.63  | 8.18  | 9.52  | 10.12 |
|                                                                   | HDD50       | 8550   | 1724  | 1374  | 1012  | 537   | 274   | 32    | 6     | 27    | 244   | 659   | 1119  | 1542  |
|                                                                   | HDD65       | 13561  | 2189  | 1794  | 1477  | 987   | 730   | 356   | 270   | 372   | 686   | 1124  | 1569  | 2007  |
|                                                                   | CDD50       | 467    | 0     | 0     | 0     | 0     | 10    | 126   | 203   | 120   | 8     | 0     | 0     | 0     |
|                                                                   | CDD65       | 1      | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0     |
|                                                                   | CDH74       | 45     | 0     | 0     | 0     | 0     | 0     | 9     | 28    | 8     | 0     | 0     | 0     | 0     |
|                                                                   | CDH80       | 1      | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0     |
|                                                                   | WSAvg       | 4.4    | 2.5   | 3.4   | 4.6   | 5.6   | 6.5   | 5.1   | 4.6   | 4.6   | 4.7   | 4.4   | 3.5   | 2.8   |
|                                                                   | PrecAvg     | 6.5    | 0.1   | 0.1   | 0.2   | 0.4   | 0.6   | 1.1   | 1.7   | 1.2   | 0.5   | 0.3   | 0.3   | 0.2   |
| Precipitation                                                     | PrecMax     | 9.4    | 0.4   | 0.2   | 0.6   | 0.7   | 1.1   | 3.1   | 4.3   | 2.6   | 0.9   | 0.6   | 0.5   | 0.4   |
|                                                                   | PrecMin     | 4.4    | 0.0   | 0.0   | 0.0   | 0.2   | 0.2   | 0.3   | 0.5   | 0.3   | 0.2   | 0.0   | 0.1   | 0.0   |
|                                                                   | PrecStd     | 1.3    | 0.1   | 0.0   | 0.2   | 0.2   | 0.3   | 0.6   | 0.9   | 0.6   | 0.2   | 0.2   | 0.1   | 0.1   |
|                                                                   | PrecStd     | 1.3    | 0.1   | 0.0   | 0.2   | 0.2   | 0.3   | 0.6   | 0.9   | 0.6   | 0.2   | 0.2   | 0.1   | 0.1   |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% DB     | 26.3   | 28.8  | 44.4  | 57.6  | 67.2  | 75.3  | 77.1  | 74.9  | 68.3  | 54.7  | 41.8  | 26.1  |       |
|                                                                   | 0.4% MCWB   | 22.4   | 23.3  | 33.2  | 41.8  | 47.5  | 54.4  | 55.0  | 52.3  | 47.7  | 40.4  | 33.0  | 21.0  |       |
|                                                                   | 2% DB       | 18.7   | 24.6  | 40.4  | 53.4  | 62.6  | 71.1  | 73.6  | 71.3  | 63.3  | 50.0  | 36.9  | 22.0  |       |
|                                                                   | 2% MCWB     | 15.4   | 19.8  | 30.2  | 39.1  | 46.3  | 52.3  | 54.2  | 51.1  | 46.3  | 37.8  | 30.3  | 18.4  |       |
|                                                                   | 5% DB       | 15.5   | 21.3  | 36.9  | 49.6  | 59.1  | 68.0  | 70.9  | 68.4  | 59.6  | 46.1  | 32.0  | 19.1  |       |
|                                                                   | 5% MCWB     | 12.8   | 17.4  | 28.5  | 37.1  | 44.4  | 50.8  | 52.9  | 49.8  | 44.9  | 35.9  | 26.7  | 16.0  |       |
|                                                                   | 10% DB      | 11.5   | 18.1  | 33.6  | 46.1  | 55.5  | 65.1  | 68.0  | 65.6  | 56.3  | 42.0  | 27.6  | 16.1  |       |
|                                                                   | 10% MCWB    | 9.1    | 14.6  | 26.7  | 35.6  | 42.0  | 49.7  | 51.7  | 49.3  | 43.3  | 33.5  | 23.1  | 13.2  |       |
|                                                                   | 0.4% WB     | 23.0   | 23.6  | 34.0  | 44.3  | 50.7  | 57.4  | 59.6  | 56.7  | 51.3  | 41.9  | 33.7  | 21.9  |       |
|                                                                   | 0.4% MCDB   | 25.5   | 27.6  | 41.4  | 55.0  | 64.6  | 71.1  | 72.1  | 68.6  | 62.9  | 51.3  | 40.3  | 25.5  |       |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2% WB       | 15.9   | 20.2  | 31.2  | 40.9  | 47.5  | 54.3  | 56.9  | 54.0  | 48.3  | 39.2  | 30.6  | 18.7  |       |
|                                                                   | 2% MCDB     | 18.5   | 24.0  | 39.0  | 49.8  | 59.4  | 66.8  | 68.8  | 65.8  | 59.8  | 48.7  | 36.7  | 22.0  |       |
|                                                                   | 5% WB       | 12.6   | 17.6  | 28.9  | 38.4  | 45.3  | 52.7  | 55.3  | 52.2  | 46.1  | 36.6  | 27.0  | 16.0  |       |
|                                                                   | 5% MCDB     | 15.4   | 21.4  | 36.1  | 47.7  | 57.2  | 64.5  | 66.4  | 64.0  | 57.5  | 45.1  | 32.0  | 19.0  |       |
|                                                                   | 10% WB      | 9.2    | 14.6  | 26.8  | 36.1  | 43.3  | 51.1  | 53.5  | 50.6  | 43.9  | 34.1  | 23.3  | 13.3  |       |
|                                                                   | 10% MCDB    | 11.5   | 18.0  | 33.4  | 45.5  | 54.4  | 62.2  | 64.4  | 62.3  | 55.1  | 41.5  | 27.6  | 16.0  |       |
| Mean Daily Temperature Range                                      | MDBR        | 20.7   | 22.8  | 23.8  | 20.3  | 18.3  | 18.8  | 18.7  | 21.3  | 21.7  | 19.3  | 19.2  | 18.8  |       |
|                                                                   | 5% DB       | 21.6   | 25.2  | 26.1  | 26.1  | 24.3  | 23.7  | 24.2  | 26.9  | 28.7  | 26.1  | 24.1  | 20.1  |       |
|                                                                   | 5% WB       | 18.8   | 21.4  | 18.3  | 15.3  | 12.8  | 9.6   | 9.6   | 11.2  | 14.8  | 15.8  | 17.9  | 17.3  |       |
|                                                                   | MCDBR       | 21.0   | 24.5  | 22.8  | 22.4  | 21.9  | 20.5  | 21.1  | 22.2  | 26.0  | 24.4  | 23.1  | 19.5  |       |
|                                                                   | MCWBR       | 18.5   | 20.9  | 16.4  | 14.5  | 12.7  | 9.3   | 9.9   | 11.0  | 14.9  | 15.6  | 17.4  | 16.9  |       |
|                                                                   | MCWBR       | 18.5   | 20.9  | 16.4  | 14.5  | 12.7  | 9.3   | 9.9   | 11.0  | 14.9  | 15.6  | 17.4  | 16.9  |       |
| Clear-Sky Solar Irradiance                                        | taub        | 0.222  | 0.249 | 0.294 | 0.340 | 0.334 | 0.342 | 0.336 | 0.317 | 0.283 | 0.266 | 0.242 | 0.224 |       |
|                                                                   | taud        | 2.337  | 2.261 | 2.183 | 2.194 | 2.288 | 2.357 | 2.398 | 2.426 | 2.481 | 2.446 | 2.377 | 2.319 |       |
|                                                                   | Ebn at Noon | 280    | 292   | 292   | 288   | 293   | 291   | 291   | 293   | 295   | 284   | 267   | 262   |       |
|                                                                   | Edh at Noon | 21     | 29    | 38    | 42    | 40    | 38    | 36    | 33    | 28    | 24    | 20    | 19    |       |
| All-Sky Solar Radiation                                           | RadAvg      | 543    | 887   | 1355  | 1740  | 2033  | 2093  | 2038  | 1822  | 1479  | 933   | 557   | 423   |       |
|                                                                   | RadStd      | 22     | 32    | 67    | 77    | 130   | 113   | 117   | 69    | 69    | 68    | 39    | 27    |       |

## Historical Trends

|                             | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only           | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (6 neighbors) | N/A   | N/A     | N/A | N/A    | N/A     | -1.13 | -1.80 | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page